REKORA

FROM IDEA TO PROTOTYPE.
FROM PROTOTYPE TO EVOLUTION.

PROJECT PROPOSAL

Integrated Engineering Project Development & Prototyping Ecosystem

SUBMITTED TO

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SUBMITTED BY

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On behalf of the REKORA Founding Team

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01. EXECUTIVE SUMMARY

> Engineering projects should not end at submission.

Today's student engineering projects involve a highly disconnected set of activities. A typical engineering student must independently navigate project discovery, component sourcing, CAD modification, fabrication, coding, testing, and documentation. Once a project is evaluated, the physical hardware is often discarded, and the engineering knowledge developed during the process is lost.

REKORA connects these fragmented activities into one continuous ecosystem.



> **REKORA transforms an individual student project into a reusable, continuously evolving engineering asset.**

To validate this model, we propose an initial controlled pilot at St. Joseph's College of Engineering. This pilot introduces component rental to drastically lower student costs, direct campus-based 3D printing, and peer-to-peer resource exchange. We seek an initial seed allocation of ₹1,00,000–₹2,00,000 to establish the hardware inventory and deployment operations.

02. THE PROBLEM

The academic engineering lifecycle is economically inefficient and structurally fragmented.

01

FRAGMENTED RESOURCES

Ideation, component sourcing, CAD modification, manufacturing, and documentation exist in isolated silos.

02

REPEATED HARDWARE PURCHASES

Students frequently purchase expensive components and hardware kits required only temporarily for a single prototype.

03

LIMITED ACCESS TO FABRICATION

Moving from digital CAD to a physical prototype is a bottleneck due to limited accessible 3D printing facilities.

04

LOSS OF PROJECT KNOWLEDGE

Once evaluated, troubleshooting steps and design logic disappear. There is no centralized system to capture output.

05

PROJECTS RESTART FROM ZERO

Because prior work is inaccessible, students cannot easily build upon previous engineering solutions.

06

LIMITED CONTINUITY

Subsequent cohorts are forced to build identical basic systems year after year, preventing ambitious engineering evolution.

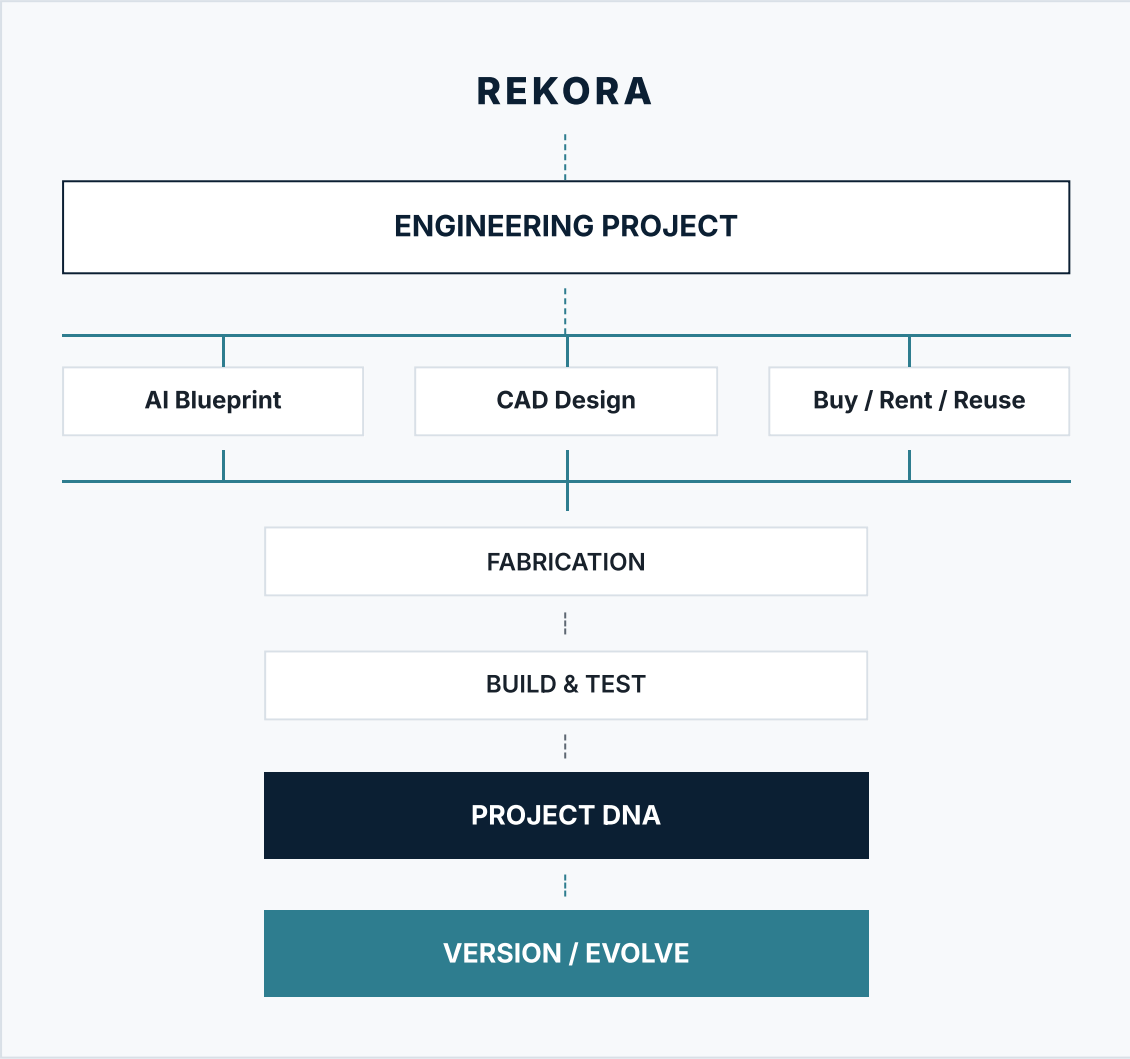
03. THE OPPORTUNITY

By consolidating the fragmented components of project development into a unified platform, we can reduce the financial barrier to innovation while dramatically improving the quality of student output.

> The integration of project discovery, hardware sourcing, and digital repositories opens the door for a circular academic economy—where resources are rented, reused, and continuously iterated upon.

04. REKORA — THE INTEGRATED ECOSYSTEM

REKORA is designed to guide an engineering project from conceptualization to physical realization, and finally to archival and evolution.



05. HOW REKORA WORKS

DISCOVERY & AI

Users begin with a problem, budget, or components. The platform algorithmically recommends suitable initiatives, while the AI Project Engineer frames constraints and generates a blueprint.

COMPONENT SOURCING

To reduce unnecessary purchases, REKORA integrates project BOMs directly with inventory. Students can choose to Buy, Rent, Use Existing, or Borrow components item by item.

CAD & FABRICATION

REKORA connects digital design to the physical build. Students customize CAD models aligned with their requirements, routing them directly to campus 3D printing.

BUILD & DOCUMENTATION

During physical assembly, REKORA provides integration logic, troubleshooting, and standardized documentation frameworks to support academic submission rigor.

PROJECT DNA

Every completed project receives a persistent digital identity preserving the exact configuration of hardware, software, and logic for future reference.

EVOLUTION & MARKETPLACE

Future users fork the repository to improve the physical design or code, while senior cohorts trade unused hardware via a peer-to-peer exchange.

06. PROJECT PREVIEW & UNLOCK MODEL

When a user discovers a project, they do not immediately receive the complete implementation logic. The platform uses a Preview & Unlock model to protect IP and ensure suitability.

AUTONOMOUS FLOATING WASTE COLLECTOR

Robotics • IoT • Computer Vision

DIFFICULTY	EST. BUILD COST	EST. TIME
Intermediate	₹4,500–₹6,000	2–4 Weeks

ABSTRACT

A low-cost autonomous surface vessel designed to identify and collect macro-plastics in static water bodies using localized computer vision.

KEY CAPABILITIES

- Autonomous navigation
- Waste detection via CV
- Modular collection nets

TECHNOLOGIES

ESP32 | Camera | GPS | Motor Control

> **WHAT YOU GET UPON UNLOCK:**
>
> Full Architecture • BOM • CAD Files • Source Code • Testing Plans • Docs

UNLOCK PROJECT REPOSITORY

THE PREVIEW ANSWERS:
What? Why? Who is it for? How difficult? What does it use?

THE REPOSITORY ANSWERS:
How exactly?

07. PROJECT DNA & VERSION EVOLUTION

Once unlocked, the Project Repository provides a structured environment for execution. More importantly, every completed project retains its engineering history as **Project DNA**.

This version control philosophy allows future students to study, build, modify, test, and contribute to an existing project rather than discarding it. The project becomes a continuously evolving engineering asset.

PROJECT REPOSITORY

01 OVERVIEW
02 REQUIREMENTS
03 HARDWARE (BOM)
04 SOFTWARE (Code)
05 CAD & DESIGN
06 FABRICATION
07 BUILD INSTRUCTIONS
08 TESTING RESULTS
09 DOCUMENTATION
10 VERSION HISTORY

THE EVOLUTION TREE CONCEPT

PROJECT: Autonomous Waste Collector

- **v1.0 (Initial Prototype)**
- **v1.1**
  - Improved chassis design
- **v1.2**
  - Reduced BOM cost via alternate motors
- **v2.0 (Major Evolution)**
  - Integrated AI vision processing
  - Full autonomous navigation capability

08. COMPONENT BUY / RENT / REUSE

REKORA connects project BOM items directly with campus inventory availability, significantly reducing the financial burden of hardware acquisition.

PROJECT COMPONENT	INVENTORY STATUS	ACTION OPTIONS
ESP32 Microcontroller	Available in Inventory	**[BUY]** or **[RENT]**
Motor Driver (L298N)	User already owns	**[USE EXISTING]**
Custom Sensor Housing	Digital CAD file only	**[3D PRINT]**
Standard Bearing 608ZZ	Consumable	**[BUY]**

QR-BASED COMPONENT LIFECYCLE

To support this ecosystem, physical components are managed via a QR-based lifecycle system (e.g., RK-ESP32-001). This operational infrastructure ensures quality control across the rental inventory.

01 **IDENTIFY** → 02 **PROJECT USE** → 03 **RETURN** → 04 **INSPECT & REUSE**

09. BUSINESS & REVENUE MODEL

REKORA establishes a diversified, sustainable revenue architecture that minimizes reliance on low-margin commodity sales by monetizing services, access, and asset utilization.

INITIAL REVENUE MODEL (CAMPUS PILOT)

REVENUE STREAM	DESCRIPTION
Project Repository Access	Users pay to unlock complete project resources, CAD, and source code.
Component Rental	Recurring high-margin utilization of physical inventory hardware.
Component Sales	Standard retail margin on consumable components and integrated project kits.
3D Printing & Fabrication	Manufacturing and service margins on direct campus-based 3D printing.
CAD & Design Services	Fees for paid design modifications and specific engineering assistance.
Marketplace Commission	Percentage commission on peer-to-peer transactions between students.

FUTURE REVENUE OPPORTUNITIES (SCALING)

REVENUE STREAM	DESCRIPTION
Technical Assistance	Premium troubleshooting, mentoring, and structured engineering support tiers.
Premium Platform Features	Subscription model for advanced AI Project Engineer capabilities.
Institutional Services	College-level SaaS subscriptions for centralized campus project archives.
Industry Sponsorships	Monetizing corporate-sponsored engineering challenges and prototype development.

10. INITIAL COLLEGE PILOT & BUDGET

ST. JOSEPH'S COLLEGE OF ENGINEERING

Initial Controlled Pilot Environment

The objective of this college-first approach is to rigorously validate demand, operations, and willingness-to-pay before scaling. The pilot follows a strict iterative model:

BUILD → VALIDATE → REVENUE → REINVEST → EXPAND

INITIAL SEED REQUIREMENT

₹1,00,000 – ₹2,00,000

CAPITAL ALLOCATION CATEGORY	ESTIMATED RANGE (INR)
Initial Component Inventory Microcontrollers, sensors, robotics hardware	₹50,000 – ₹80,000
3D Printing & Fabrication Materials and equipment maintenance	₹20,000 – ₹40,000
QR & Inventory Tracking Physical storage, labeling systems	₹5,000 – ₹10,000
Operations & Cloud Logistics Packaging, server hosting, API overhead	₹10,000 – ₹20,000
Marketing & Launch Student awareness, onboarding	₹5,000 – ₹10,000
Contingency Reserve	₹10,000 – ₹20,000
Reference Total Allocation	**₹1,00,000 – ₹1,80,000**

11. RISK & MITIGATION

RISK FACTOR	MITIGATION STRATEGY
Inventory lock-in	Start exclusively with highly standardized, high-utilization components.
Hardware damage	Implement QR tracking, security deposits, and mandatory inspection.
Academic integrity	Strictly prohibit ready-made submissions; focus solely on resources.
Operational complexity	Restrict rollout entirely to a controlled college pilot before any expansion.

12. FOUNDING TEAM

REKORA is driven by a specialized team possessing the technical and operational expertise necessary to execute this infrastructure.

Reshma S
FOUNDER & CHIEF EXECUTIVE OFFICER

Strategic vision, enterprise growth, and institutional partnerships.

Rogith Kumar R
CO-FOUNDER & CHIEF OPERATING OFFICER

Operational scaling, execution strategy, and continuous process optimization.

NandhaKumar G
CO-FOUNDER & CHIEF TECHNOLOGY OFFICER

Lead software development, additive manufacturing, and systems integration.

Monica S
HEAD OF SUPPLY CHAIN & LOGISTICS

Asset lifecycle management, inventory control, and precision procurement tracking.

Justin I
LEAD PRODUCT ENGINEER

Parametric modeling, rapid prototyping, and physical hardware architecture.

Sanjay B
HEAD OF STRATEGIC PARTNERSHIPS

Stakeholder engagement, requirement analysis, and academic relations.

REKORA

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Supporting REKORA is an opportunity to establish a cumulative engineering ecosystem in which **today's project becomes tomorrow's foundation.**

REQUEST FOR INSTITUTIONAL APPROVAL & INITIAL SEED SUPPORT

₹1,00,000 – ₹2,00,000

St. Joseph's College of Engineering
Chennai
